

Advances in Sparse Bayesian Factor Models

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- we have a p -dimensional continuous (binary) vector measuring the abundance (presence) of p species of birds
- observations are grouped in $S = 200$ different locations, with $s = 1, \dots, S$.
- In each location we have repeated sampling campaigns for a total of n observation in total ($i = 1, \dots, n$)
- data y_{is} is the p -dimensional measurement for location s at sampling campaign i .

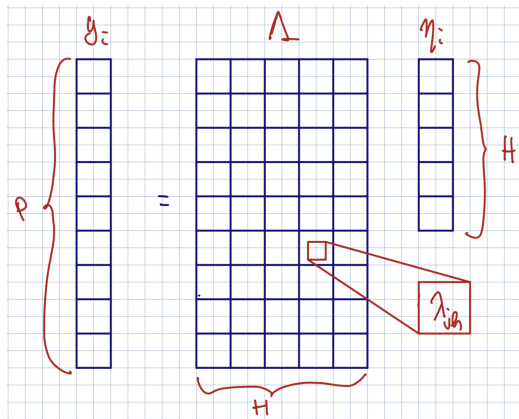
- we consider this data in a “multi-study” fashion
- other examples:
 - medical studies performed in different hospitals
 - genomics studies performed with different technological platforms
 - ...
- we are going to follow a factor model approach, and specifically a **Multi-Study Factor Analysis** (MSFA, De Vito et al. 2017) approach

- 1 Introduction
- 2 Adaptive partition factor analysis
 - From MSFA to APAFA
 - Empirical evidence
- 3 Identifiable Bayesian FM via Sparse Latent Factors
 - Geometric identification
 - Statistical (asymptotic) identification
- 4 Conclusions

Factor Analysis (FA)

$$y_i = \Lambda \eta_i + \epsilon_i \quad \epsilon_i \sim f_\epsilon, \quad \eta_i \sim f_\eta,$$

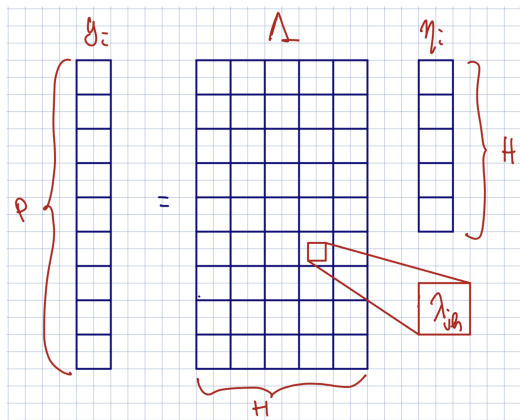
- y_i : i -th p -variate random variable;
- Λ : $p \times H$ factor loadings matrix;
- η_i : i -th vector of H latent factors.



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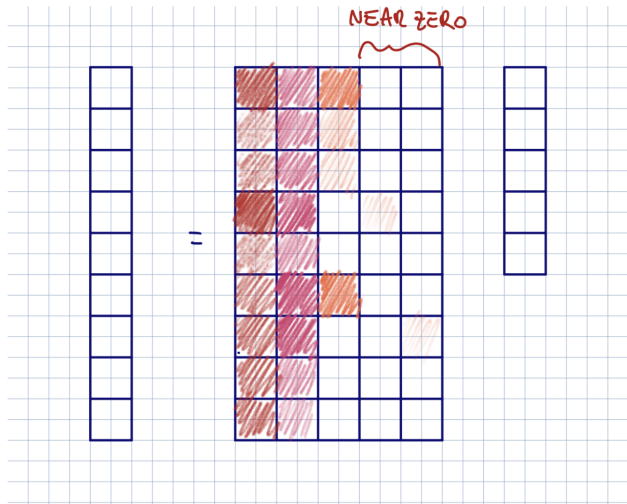


If $\epsilon_i \sim N(0, \Sigma)$, $\eta_i \sim N(0, I)$, then marginally

$$y_i \sim N(0, \Lambda \Lambda^\top + \Sigma)$$

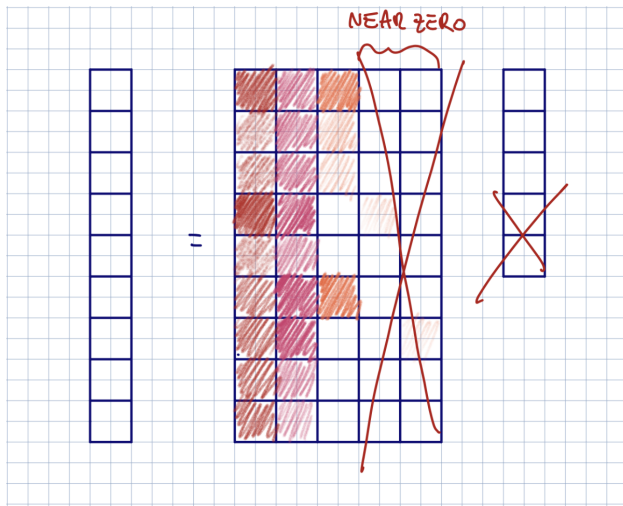
- Most of the interest of FA revolves around the concept of **interpretability**;
- Interpretation of factor models is assigning a meaning to the latent factors and then to their impact on the observed data;
- this is promoted by the concept of **sparsity** in many ways...

Sparsity for dimensionality reduction



Interpretation of loadings matrix and factors is strongly favoured by a limited number of factors

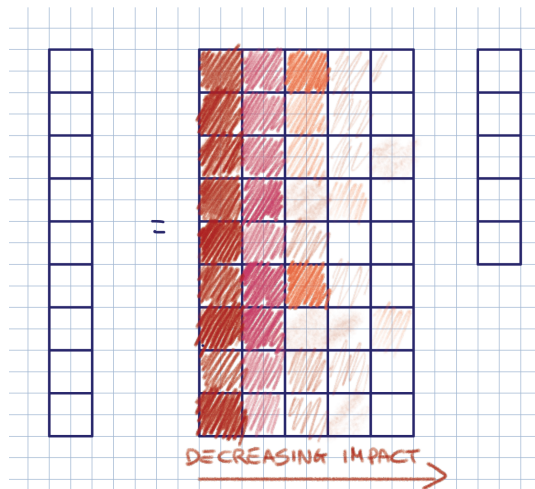
Sparsity for dimensionality reduction



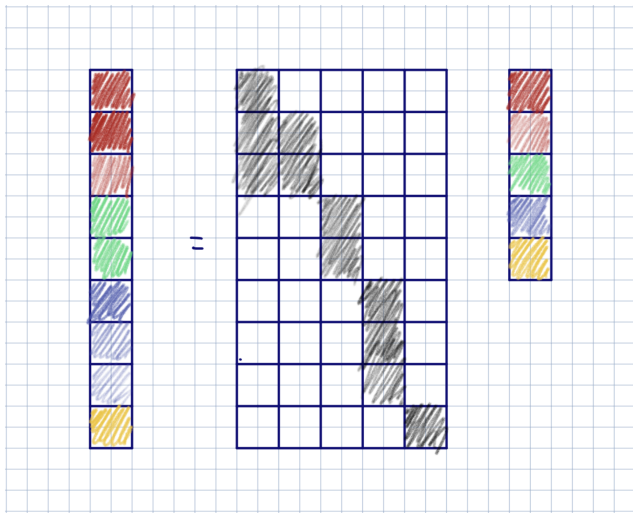
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Bayesian (infinite) factor models

- Infinite factor models are Bayesian Nonparametric models for dimensionality reduction
- Batthacharya & Dunson (2011) first introduced this idea with the **multiplicative gamma process** (MGP)
- Similar in spirit is the **cumulative shrinkage process** (CUSP) by Legramanti et al. (2020) recently generalized by Frühwirth-Schnatter (2024)



Sparsity for block structure



Interpretation of loadings matrix is strongly favored when each factor has an impact only on a small group of components of observed variables

- Local sparsity is another key concept in Bayesian FA
- Priors for each λ_{jh} can be broadly divided into
 - Continuous shrinkage prior: Zhao et al. (2016), Rockova & George (2017), Kastner (2019)
 - Discrete shrinkage priors (spike & slab): West (2003), Carvalho et al. (2008), Conti et al. (2014), Kaufmann & Schumacher (2019), Schiavon et al. (2022), and Frühwirth-Schnatter et al. (2025)

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- Specifically

$$y_{is} = \Lambda \eta_{is} + \Gamma_s \varphi_{is} + \epsilon_{is} \quad (1)$$

where Γ_s is a (study-specific) factor loading matrix of dimension $p \times k_s$, with $k_s \ll p$ possibly different in each study, and φ_{is} its corresponding latent factor.

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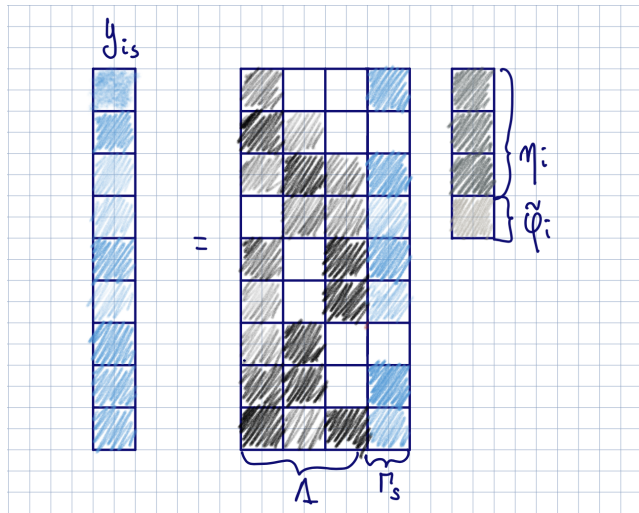
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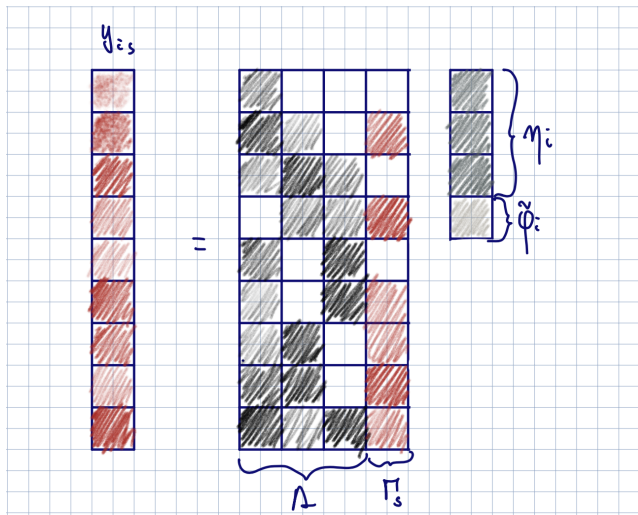
- For Gaussian noise, the marginal distribution of y_{is} is Gaussian with covariance

$$\Omega_s = \Lambda \Lambda^\top + \Gamma_s \Gamma_s^\top + \Sigma$$

MSFA graphically



MSFA graphically

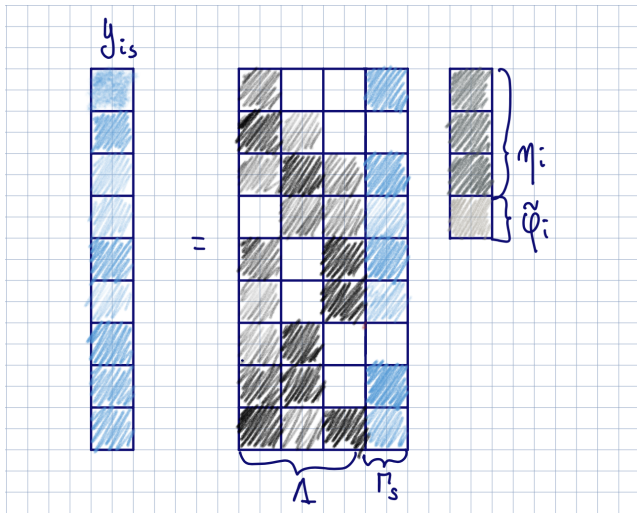


- Rewrite the MSFA as

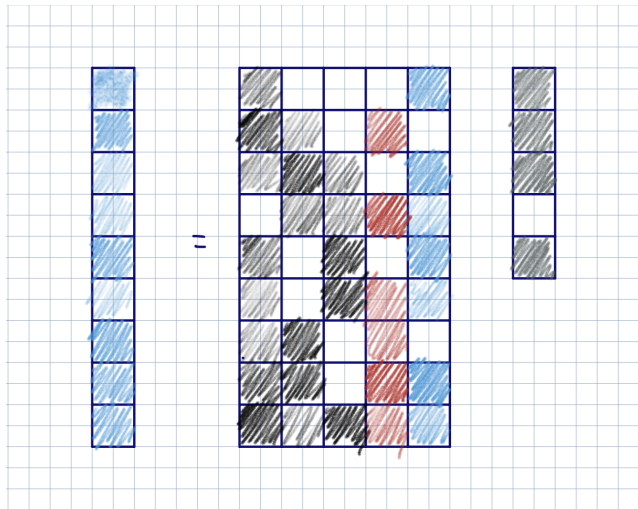
$$y_i = \Lambda \eta_i + \Gamma \varphi_i + \epsilon_i. \quad (2)$$

- Here $\Gamma = (\Gamma_1, \dots, \Gamma_S)$ concatenate along columns all the study-specific factor loading matrices into a $p \times k$ matrix with $k = \sum_{s=1}^S k_s$
- φ_i is a k -dimensional augmented vector containing the original φ_{is} framed with suitable pattern of zeroes.

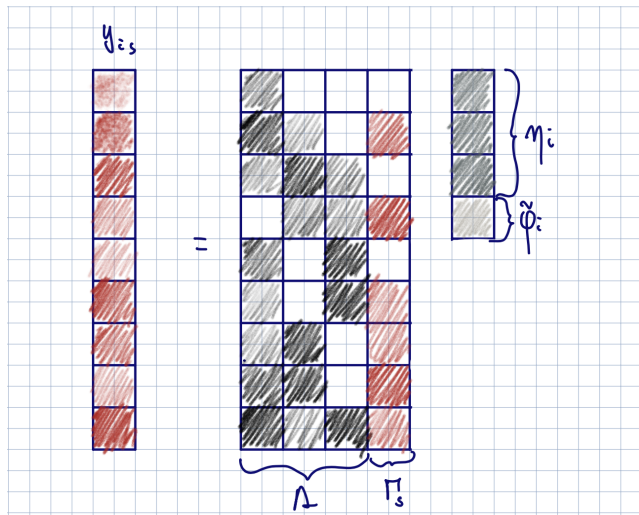
Rethinking the MSFA graphically



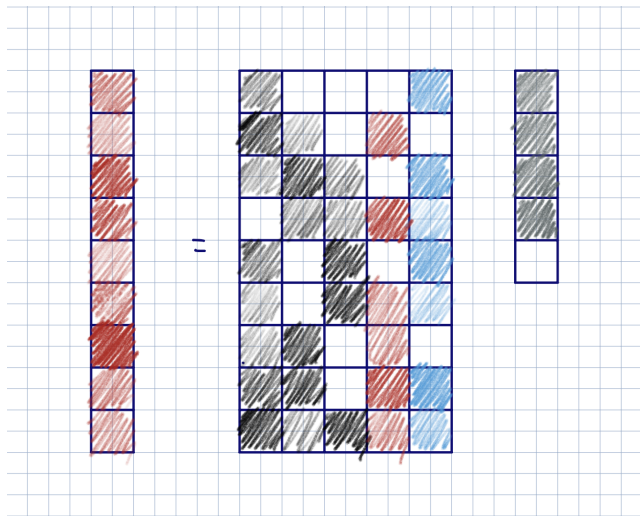
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Rethinking the MSFA graphically



Rethinking the MSFA graphically



Fixed partition vs Adaptive partition

- MSFA permits precisely S study-specific loading matrices Γ_s
- practical scenarios often present more complex situations:
 - two or more studies may present high homogeneity, potentially sharing identical or nearly identical latent representations
 - some studies may involve a highly heterogeneous group of subjects, possibly leading to two or more sub-populations displaying distinct latent representations
- An **adaptive** partition that accounts for the above situations (and beyond) would be useful!

- Marginalizing out the (shared) latent factors $\eta_i \sim N(0, I_p)$,

$$y_i \sim N(\Gamma \varphi_i, \Lambda \Lambda^\top + \Sigma). \quad (3)$$

- Let x_i be a dummy variable scharacterizing the study to which unit i belongs
- The h -th element of the vector φ_i is $\varphi_{ih} = \tilde{\varphi}_{ih} \psi_{ih}$ where $\tilde{\varphi}_{ih}$ is a continuous random variable and $\psi_{ih} = f_h(x_i)$ with f_h a deterministic activation function.
- for MSFA $f_h(x_i) = x_i^\top 1_S$ where 1_S is a S dimensional vector of ones

- What about incorporating the information contained in the x_i 's in a more flexible manner?
- For example with

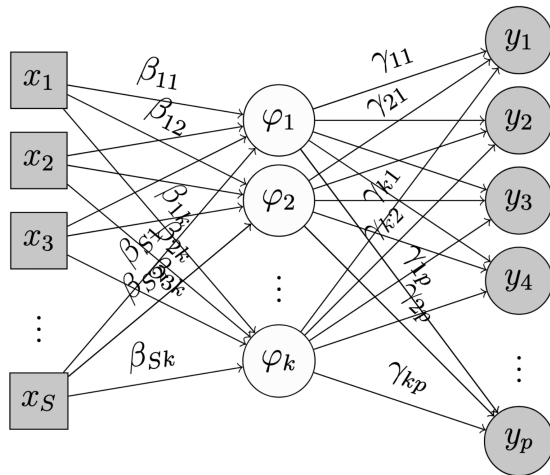
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- Which leads to a specific single layer neural network where
 - the x_i s are the input variables,
 - y_i are output variables,
 - φ_i s are the nodes of the hidden layer
 - f_h are the activation functions
 - β_h are the weights between the input and the hidden layer,
 - Γ are the weights between the hidden layer and the outcome

From factor analysis to neural networks



- We encode the study-specific membership into categorical the variables x_i
- The study-specific latent factors are assumed

$$\varphi_{ih} \sim N(0, \psi_{ih}(x_i)\tau_h).$$

- We then assume a dependence between the scale parameters of the group-specific latent factors and the group indicator x_i , as follows:

$$\psi_{ih}(x_i) \sim \text{Ber}\{\text{logit}^{-1}(x_i^\top \beta)\}.$$

- The shrinkage prior on the elements $\varphi_{ih}(x_i)$ enables and promotes, yet does not mandate, the sparse representation of MSFA
- Wide range of scenarios including:
 - two or more studies exhibit high homogeneity and share nearly identical latent representations,
 - some studies involve highly heterogeneous groups of subjects, potentially resulting in two or more sub-populations with distinct latent structures,
 - any combinations of the above.

We simulate data under the following scenarios

- *Scenario A — Correct specification:* $S = 3$ groups with sample size $n_1 = n_2 = n_3$, $d = 2$ active shared factors, and $k = 3$ ($[1 + 1 + 1]$ for the groups) specific factors
- *Scenario A* — Latent Heterogeneity:* we do not provide the group labels but the data are generated as in Scenario A.
- *Scenario B — Homogeneity between groups:* While we provide $S = 3$ groups, the structure of latent factors is homogeneous among all the studies, i.e. $k = 0$.
- *Scenarios C and D — Mixed situations:* There exist groups but $k \neq S$



(A, A*)

(B)

(C)

(D)

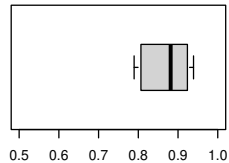
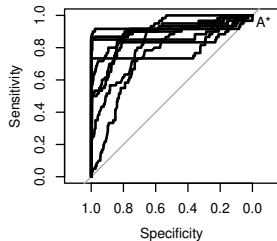
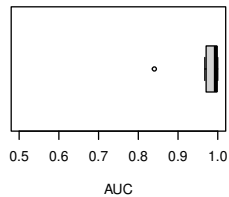
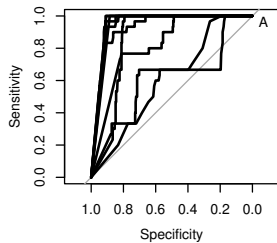
- We evaluate the performance of **APAFA** with the approach proposed Gabski et al., 2023 (**TETRIS**)
- For our method only, we evaluate the **ability in discovering the group structure**.
- To compare the relative performance of each competitor, we measure the adequacy of the reconstructed the variances of each group.

Results: variance matrices reconstruction

Table: Monte Carlo average (and interquartile range) of the posterior mean number of factors and RV coefficients for Ω_1 , Ω_2 , and Ω_3 on several simulation scenarios (the higher the better).

Scen	Method	d	k	Ω_1	Ω_2	Ω_3
A	APAFA	3.00 (1.00)	3.01 (0.23)	0.90 (0.07)	0.88 (0.07)	0.89 (0.05)
	TETRIS	3.67 (1.26)	4.91 (0.89)	0.92 (0.07)	0.93 (0.09)	0.88 (0.08)
A*	APAFA	4.00 (1.00)	3.00 (1.00)	0.69 (0.13)	0.78 (0.08)	0.75 (0.08)
B	APAFA	3.00 (0.00)	0.00 (0.00)	0.94 (0.04)	0.94 (0.04)	0.94 (0.04)
	TETRIS	3.00 (0.00)	0.00 (0.00)	0.90 (0.12)	0.92 (0.05)	0.92 (0.09)
C	APAFA	3.00 (0.00)	3.00 (0.05)	0.89 (0.05)	0.78 (0.04)	0.92 (0.02)
	TETRIS	3.00 (1.00)	2.00 (1.01)	0.72 (0.09)	0.75 (0.08)	0.79 (0.05)
D	APAFA	3.00 (0.00)	3.00 (0.12)	0.91 (0.03)	0.88 (0.03)	0.90 (0.04)

Results: group discovering



- The locations are considered as **groups in a multi-study framework**
- **y** : $n \times p$ binary matrix of occurrence of **p species** in **n** different **sampling campaigns**.
- We model species presence or absence using the multivariate probit regression model,

$$y_{ij} = 1(z_{ij} > 0), \quad z_i = \Lambda \eta_i + \Gamma \varphi_i + \epsilon_i, \quad \epsilon_i \sim N(0, \Sigma).$$

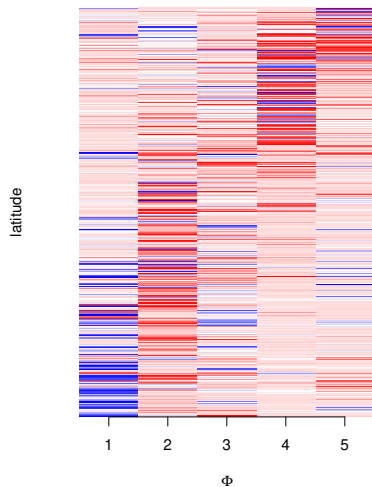
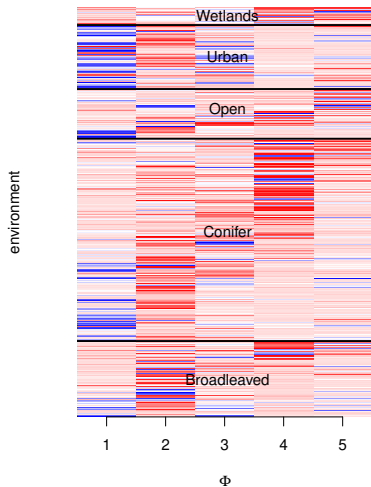
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- **$S = 200$** : number of sites
- we do not use any information but the sampling campaign indicator. However ...
- The 200 locations can be clustered into 5 different types of location: Urban, Broadleaved forests, Coniferous forests, Open, and Wetlands.

Posterior estimate of specific factors



- Factor loadings are not identifiable due to rotations, i.e .

$$\Lambda\Lambda^T = \tilde{\Lambda}\tilde{\Lambda}^T, \quad \text{if} \quad \tilde{\Lambda} = \Lambda P, \quad \text{and} \quad P \in \mathcal{O}_K$$

Hence the likelihood identifies the *orbit*

$$\mathcal{O}(\Lambda) = \{\Lambda P : P \in \mathcal{O}_K\},$$

but not a specific orientation of Λ .

- In the multi-study context and, specifically, in

$$y_i \sim N(0, \Omega_i), \quad \Omega_i = \Lambda\Lambda^T + \Gamma \text{diag}\{\psi_i\}\Gamma^T + \Sigma$$

we may additionally face “information switching”

Definition (Information Switching)

Let S_n the number of distinct groups i.e. $S_n = |\cup_{i=1}^n \Omega_i|$. Denote with Ψ the $n \times k$ matrix that stacks in distinct rows all $\psi_i = (\psi_{i1}, \dots, \psi_{ik})$ and with Ψ_h its generic column with, $\Psi_h \neq \mathbf{1}_n$ for all $h = 1, \dots, k$.

Let Ω_s^* and ψ_s^* ($s = 1, \dots, S_n$) be the distinct values of Ω_i and ψ_i , respectively.

Similarly, let $W_s^* = \Omega_s^* - \Lambda\Lambda^\top - \Sigma = \Gamma \text{diag}\{\psi_s^*\} \Gamma^\top$.

The model suffers from information switching if there exist $\tilde{\Gamma} \neq \Gamma$ and $\tilde{\Psi} \neq \Psi$ such that $W_s^* = \tilde{\Gamma} \text{diag}\{\tilde{\psi}_s^*\} \tilde{\Gamma}$ for all s , with $\tilde{\Psi}_h = \mathbf{1}_n$ for at least one h .

Theorem

If $\Psi_h \neq \mathbf{1}_n$ for all $h \in \{1, \dots, k\}$ and Γ is of full column rank k with $k < p(p+1)/2$, then the model is resistant to information switching.

Definition (Non-replicable Sparsity Pattern Condition)

All columns of Ψ^* , where Ψ^* is the $S_n \times k$ matrix stacking all the distinct ψ_s^* , are different.

Theorem

Let $\Gamma \in \mathbb{R}^{p \times k}$ be a real matrix with full column rank k . If $P \in \mathcal{O}_k$ and $\Gamma' = \Gamma P$ is a rotation of the specific factors, under the Non-replicable Sparsity Pattern Condition, then

$$\Lambda \Lambda^\top + \Gamma \Gamma^\top + \Sigma = \Lambda \Lambda^\top + \Gamma' \Gamma'^\top + \Sigma,$$

and, for each $s = 1, \dots, S_n$

$$\Lambda \Lambda^\top + \Gamma \text{diag}(\psi_s^*) \Gamma^\top + \Sigma = \Lambda \Lambda^\top + \Gamma' \text{diag}(\psi_s^*) \Gamma'^\top + \Sigma,$$

if and only if P is a permutation matrix.

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 - classical approaches rely on positive lower-triangular (PLT) constraints.
 - more recent Bayesian approaches use generalized lower-triangular (GLT) structures; (Frühwirth-Schnatter, et al., 2025; Canale and Frühwirth-Schnatter, 2026).

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- An alternative is to apply post-hoc rotations to point estimates or MCMC draws (Poworoznek et al., 2026)

A geometric identifiability



- With APAFA's **sparsity** we show that rotational identifiability (of the *specific* part) is associated with the **latent factors** sparsity.
- Note: latent factors (scores) not factor loading!

- With APAFA's **sparsity** we show that rotational identifiability (of the *specific* part) is associated with the **latent factors** sparsity.
- Note: latent factors (scores) not factor loading!
- Consider the general setting (no multi-study),

$$y_i = \Lambda \eta_i + \varepsilon_i, \quad \eta_i = \psi_i \odot \tilde{\eta}_i, \quad \tilde{\eta}_i \sim N_K(\mathbf{0}, I_K),$$

where

$$\psi_i \in \{0, 1\}^K$$

selects the active factors for unit i .

- Conditionally on ψ_i ,

$$y_i \mid \psi_i \sim N_p(\mathbf{0}, \Lambda \text{diag}(\psi_i) \Lambda^T + \Sigma).$$

A structured scale mixture of Gaussians



If the sparsity patterns are random, then marginally

$$y_i \sim \sum_{\psi \in \{0,1\}^K} \Pr(\psi_i = \psi) N_p(0, \Lambda \text{diag}(\psi) \Lambda^T + \Sigma).$$

This is a structured covariance mixture:

$$\Sigma_\psi = \Omega \text{diag}(\psi) \Lambda^T + \Sigma.$$

Unlike a generic mixture of factor analyzers, all components share the same global loading matrix Λ .

Interpretation

Mixture components correspond to nested or partially overlapping factor subspaces of a common latent structure.

The simplest identifying pattern: anti-pivots



Among all non-replicable sparsity structures, the simplest one uses one factor-specific zero per factor.

A unit i is an *anti-pivot* for factor k if

$$\psi_i = \mathbf{1}_K - e_k.$$

Then

$$\Omega_k = \Lambda(I_K - e_k e_k^T) \Lambda^T + \Sigma = \Omega - \lambda_k \lambda_k^T.$$

Therefore,

$$\Omega - \Omega_k = \lambda_k \lambda_k^T.$$

Consequence

Each anti-pivot covariance isolates one rank-one loading contribution. Thus the loading columns are identified up to signs.

Anti-pivot structure



	1	2	3	4	K
1	0	1	1	1	1
2	1	0	1	1	1
$\vdots$	1	1	0	1	1
i	1	1	1	0	1
$\vdots$	1	1	1	1	1
n	1	1	1	1	1

at least one
structural zero per factor

- We exploit this geometric property to achieve identification through sparsity in the latent factors.
- We characterize differences with classical Gaussian factor models.
- ...
- This provides an alternative route to identification that does not require structural constraints on the loading matrix.

Asymptotics: the anchoring trade-off



Posterior consistency

Is the posterior concentrating on a neighborhood of the true factor loading for $n \rightarrow \infty$?

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Let m_n be the minimum number of anti-pivots per factor. In practice we need

$$m_n \rightarrow \infty \quad \text{but} \quad \frac{K_n m_n}{n} \rightarrow 0.$$

Thus, the amount of information anchoring each factor orientation can diverge, while the fraction of observations departing from the classical Gaussian factor model vanishes.

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Interpretation

A small, asymptotically negligible subset of structured observations can identify and estimate the loading orientation.

Posterior contraction of the identified loadings



Let

$$\rho_n = c_n \sqrt{\frac{s_n K_n \log p_n}{m_n}}, \quad b_n = \min_{1 \leq k \leq K_n} \|\lambda_{0n,k}\|_2^2.$$

Assume

$$\frac{\rho_n}{b_n} \rightarrow 0.$$

Then, for sufficiently large M ,

$$\mathbb{E}_{0n} \Pi_n \left\{ d_{\max, \pm}(\Lambda, \Lambda_{0n}) > M \frac{\rho_n}{\sqrt{b_n}} \mid y^{(n)} \right\} \rightarrow 0.$$

Take-home

Anti-pivots upgrade covariance contraction into loading-column contraction.

Wrapping up & essential references



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Thank you for your attention!

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